

DAYANANDA SAGAR COLLEGE OF ENGINEERING

**Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning**



INTERNSHIP REPORT

ON

“AI-Powered Evaluation and Summarization of Insurance Legal Documents using LLMs”

Submitted in partial fulfillment for the award of degree(21INT82)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Fatima Anees Mulla (1DS21AI017)

Conducted at



DocLens.ai (AI-driven InsurTech Company)

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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CERTIFICATE

This is to certify that the Internship titled “**AI-Powered Evaluation and Summarization of Insurance Legal Documents using LLMs**” carried out by **Fatima Anees Mulla [1DS21AI017]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

Signature of Reporting Head
Mr. Amit Saha

Head of AI
DocLens.ai

Signature of Guide
Prof. Anupama V P
Dept. of AIML

DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Fatima Anees Mulla**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

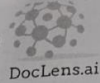
Date:

USN: 1DS21AI017

Place: Bangalore

NAME: Fatima Anees Mulla

OFFER LETTER



Fatima Anees Mulla
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Feb 14, 2025

Dear Fatima,

Congratulations! On behalf of **DocLens.ai India Private Limited (Company)**, I am pleased to offer you an internship position subject to your acceptance of the following terms and conditions of service:

1. **Position:** Information related to the start of your appointment is listed below. Your reporting manager will determine your responsibilities and duties. You understand that your responsibilities and duties may evolve over time.

Joining Date	Feb 17, 2025
Completion Date	May 17, 2025 (tentative)
Candidate Name	Fatima Anees Mulla
Manager Name	Amit Kumar Saha
Job Title	Software Engineering Intern
Job Location	DocLens.ai India Pvt. Ltd, BHIVE, JBR TechPark, Whitefield, Bangalore, Karnataka, 560066, India.

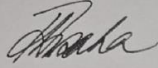
2. **Compensation:** Your stipend will be **Rs. 40,000/- per month** (Rs. Forty Thousand per month). All payments made by the Company will be in accordance with the taxation norms prescribed under the Income Tax Act, 1961.
3. **Conditions of Internship:** Your internship with the Company is subject to the completion of satisfactory background checks to ensure that you are suitable to carry out the requirements of your position, and to ensure that the information you have provided is true and correct. In the event that the background checks reveal that you have given false information or disclose unsatisfactory results, the Company reserves the right to immediately terminate your internship without notice.
4. **General Policies:**
 - a. If any information furnished by you to the Company is found to be false or inaccurate, or if you are found to have willfully suppressed any material information, the Company shall be entitled to terminate your internship without notice.
 - b. The normal operating hours for office internships are from 10.00 am to 6.00 pm, Monday through Friday. However, the work and deadlines normally decide the working hours for people.

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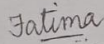
Fatima

DocLens.ai

Sincerely,
for and on behalf of **DocLens.ai India Private Limited.**



Amit Kumar Saha
Head of India
DocLens.ai India Pvt. Ltd.



Fatima Anees Mulla

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5. **Confidential Information:**

As part of your internship you may have access to Confidential Information including all information disclosed, made available or provided by or on behalf of the Company at any time that reasonably should be understood to be confidential or proprietary in light of the circumstances of such disclosure or provision, including, without limitation, information that concerns or relates to the Company's business, technology, products, methodology, coding, services, partners, consultants, customers, prospective customers, contacts, prices or performance under any agreement. Confidential Information also includes any data or information from a Company customer. Confidential Information includes any Personal Identifiable Information. You agree: (a) not to copy, disclose to third parties, or use the Confidential Information for any purpose other than to perform your work for DocLens; (b) to protect the Confidential Information with the same degree of care you exercises to protect your own confidential information (but in any event, with no less than a reasonable degree of care) to prevent the unauthorized, negligent, or inadvertent use, disclosure, or publication thereof; (c) to follow any DocLens policies or procedures relating to use or protection of Confidential Information.

6. **Termination of Internship:**

Upon the termination of your internship with the Company and prior to your departure from the Company, you agree to surrendering to the Company all proprietary or confidential information and articles that belong to the Company; and executing such documentation as is required by the Company to evidence the foregoing. Following termination of your internship you will not access or use Company data, information, tools, or property. Further, you hereby authorize the Company to offset against any final amounts that may be paid to you any amounts that you owe to the Company as of such date.

7. **Company Policies:** Upon commencement of your internship, you will be required to sign the Company's standard policies as and when required and made available to you. Your refusal and/or failure to accept the policies shall be sufficient cause for the Company to terminate your service forthwith without notice.

8. **Proprietary Rights:** You shall agree that the proprietary rights in any or all inventions, designs, applications, or work systems which you have made or developed, wholly or partially, during your internship with the Company, shall be the exclusive property of the Company.

9. **Notice:** Any notice/communication required to be served on you shall be in writing and shall be deemed to have been given on the date of delivery or transmission if personally delivered or faxed.

Please feel free to contact **Amit Saha** at amsaha@doclens.ai, if you have any questions about your appointment. You are requested to return a signed copy of the letter to the HR department for their records.

We look forward to welcoming you to **Doclens.ai India Private Limited**.

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr. Amit Saha** for guiding me during the period of internship.

We express our deep and profound gratitude to our Guide, **Prof. Anupama V P**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Fatima Anees Mulla [1DS21AI017]

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CHAPTER 1

INTRODUCTION

The growing reliance on artificial intelligence (AI) in highly specialized industries like law and insurance has created new demands for intelligent document processing, structured data extraction, and model evaluation. Legal documents, in particular, are complex, verbose, and rich in domain-specific language, requiring not only linguistic fluency but also contextual reasoning and precision in information retrieval. Recognizing these challenges, doclens.ai operates at the intersection of legal tech and AI, focusing on automating and optimizing document intelligence workflows using state-of-the-art large language models (LLMs).

The goal of my Spring 2025 internship at doclens.ai was to contribute to two critical areas: AI-driven summarization of legal case files and rubric-based evaluation of LLM-generated outputs. The first project focused on evaluating the viability of public court documents from PACER as a source for generating legally useful case summaries, in comparison to proprietary summaries from LexisNexis. This involved using advanced LLMs such as ChatGPT-4 Turbo and DeepSeek R1 to extract structured information and build summaries from unstructured legal text. The second project addressed the limitations in current evaluation frameworks by proposing a rubric-based, reasoning-driven scoring system to assess the quality and factual accuracy of AI-generated responses in insurance domain tasks.

Through this internship, I gained hands-on experience in prompt engineering, evaluation design, and comparative model analysis. I also developed skills in preprocessing legal datasets, identifying true positives and false negatives in LLM outputs, and drawing insights using structured evaluation metrics such as precision and recall. This internship not only enhanced my technical proficiency but also gave me a deeper understanding of the nuances involved in applying AI to critical legal workflows.

The experience has reaffirmed the importance of rigorous evaluation, domain adaptation, and responsible AI deployment, particularly in high-stakes fields such as law and insurance. It has also highlighted the vast potential of generative AI when combined with thoughtful design, clear rubrics, and contextual understanding.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

DocLens.ai – InsurTech Company



2.1 Introduction

Doclens.ai is a cutting-edge technology company operating at the confluence of artificial intelligence, insurance, and legal risk analytics. Founded with a vision to transform how legal and insurance professionals interact with documentation, doclens.ai leverages modern AI techniques to simplify, automate, and enhance complex workflows involving claims, litigation, compliance, and risk management.

The organization is known for building responsible, reliable, and regulatory-compliant AI products that empower users to extract meaningful insights from large volumes of unstructured legal and insurance documents. Whether it's summarizing lengthy court cases or identifying subtle risk elements within claim files, the solutions developed at doclens.ai are grounded in real-world usability, legal compliance, and enterprise-grade robustness.

The team at doclens.ai brings together over 75 years of combined experience across key industries including insurance, corporate law, AI research, software engineering, and digital transformation. Their cumulative expertise spans top 10 global insurance firms, top 3 cloud service providers, major legal entities, and global consulting giants, enabling them to architect domain-specific AI products that are technically sound and contextually intelligent.

2.2 Founders and Leadership Team

DocLens.ai was co-founded by Arnab Dey and Michael Bruton, both of whom bring decades of industry expertise to the company's mission of transforming insurance workflows through AI.

Arnab Dey – Chief Executive Officer

Arnab Dey is a visionary leader with over 25 years of experience in insurance digital transformation, analytics-driven decision-making, and startup scaling. Formerly an Executive Vice President at CapGemini and EXL Service, Arnab was also instrumental in the success of Inductis, a data analytics startup that went from inception to exit. At doclens.ai, Arnab brings strategic oversight, a customer-first mindset, and a commitment to delivering measurable impact. He holds a degree in Computer Science and an MBA, and beyond the boardroom, he is a passionate bridge player and gastronome.

Michael Bruton – Chief Product Officer

Michael Bruton is a legal and insurance law expert with over 25 years of experience working across law firms and insurance companies such as CNA. A licensed attorney, Michael brings a deep understanding of the challenges faced by legal and claims professionals, particularly those managing complex insurance litigation or risk analysis workflows. At doclens.ai, he is instrumental in product conceptualization, user experience, and regulatory alignment. Michael is also deeply involved in community service and enjoys traveling.

Amit Saha – Head of AI

Amit Saha is an accomplished technologist specializing in MLOps, AIOps, and distributed systems. With over 20 years of experience at high-impact tech companies including Cisco Meraki (Smart Cameras) and Juniper Networks(QFabric), Amit has built scalable and reliable AI systems in production. At doclens.ai, he leads the design and development of AI frameworks that power the platform's smart extraction, summarization, and evaluation capabilities. Amit is also a visiting faculty at several educational institutions in India and is passionate about badminton, photography, and travel.

2.3 Vision and Mission

Doclens.ai was created to address the growing complexity and inefficiencies in insurance and legal documentation. The company's mission is to deliver outcome-oriented AI that enhances risk identification, summarization, and information retrieval in regulated industries.

They believe that AI is not the end goal, but a powerful means to improve workflow productivity, legal accuracy, and claims analysis at scale. This vision is underpinned by the company's core values:

Responsible AI: The company prioritizes ethical design, fairness, explainability, and regulatory compliance in every AI system it builds, recognizing the high-stakes nature of the industries it serves.

Outcome-Oriented: All AI tools are designed with the end-user in mind—claims analysts, risk managers, legal professionals—ensuring that the value delivered is measurable, relevant, and immediately usable.

Innovative: doclens.ai continuously strives to stay ahead of the curve by combining cutting-edge research with practical applications. Their solutions are not only intelligent but also user-friendly and future-proof.

2.4 Products and Solutions

Doclens.ai delivers a highly specialized AI platform tailored for regulated industries like insurance and law, where documentation is dense, context-heavy, and critical to business outcomes.

At its core, the platform transforms unstructured, multi-format documents into structured, actionable insights through a suite of advanced AI tools that prioritize accuracy, explainability, and compliance.

- **Legal Document Summarization**

Legal documents—especially those from court proceedings, claims files, or case law—are notoriously long and complex. doclens.ai provides automated summarization capabilities that:

- Extract key facts such as case parties, allegations, court outcomes, and awards.
- Deliver concise, multi-paragraph summaries that preserve legal context and factual accuracy.
- Use prompt-optimized LLMs (like ChatGPT and DeepSeek) fine-tuned for legal language to ensure summaries are usable by professionals with minimal post-

processing.

- This tool enables legal analysts and researchers to rapidly triage case files, determine relevance, and reduce manual workload in document review processes.

- **Insurance Claims Risk Evaluation**

One of the platform's flagship applications is in the area of liability claims assessment—a high-impact, time-sensitive process that traditionally depends on manual expert review. doclens.ai offers tools that:

- Evaluate bodily injury, property damage, and medical cost claims using AI models trained to detect inconsistencies and omissions.
- Identify risk indicators such as missing documents, vague injuries, or inconsistent witness statements.
- Support risk scoring and adjuster triage with structured risk summaries, enabling faster settlement recommendations.
- This solution is particularly effective in high-volume claims environments, helping insurers reduce claim cycle time, legal exposure, and fraud risk while improving reserve accuracy.

- **Automated Rubric-Based Evaluation of AI Outputs**

With the widespread use of LLMs for generating legal summaries, explanations, and decisions, a new challenge arises—how do you trust what the model says? doclens.ai addresses this through:

- A proprietary rubric evaluation framework that scores AI-generated responses for accuracy, factual grounding, and reasoning completeness.
- The ability to calculate precision, recall, TP/FP/FN, and other structured metrics to evaluate model output quality.
- Integration with reasoning-capable LLMs (like DeepSeek R1) to enable explainable AI auditing.
- This solution is essential for enterprise users deploying generative AI in sensitive contexts like litigation strategy, contract review, or claims negotiation.

- **Capabilities: From Intake to Intelligence**

The doclens.ai platform supports a multi-step workflow starting from raw document ingestion to structured decision-ready insights:

- **Multi-format Document Ingestion**

Handles PDFs, scanned documents, text, tables, and even handwritten notes, applying

OCR and NLP for standardization.

- Legal-Aware Preprocessing
Removes boilerplate, deduplicates content, standardizes parties, and normalizes legal citations.
- Smart Information Extraction
AI models extract details like:
 - Court and trial details
 - Dates and timelines
 - Injury types and severity
 - Award amounts
 - Attorney details
 - Policy and claim IDs
- Contextual Summarization and Labeling
Summaries are enriched with context-aware language models that preserve factual intent and mitigate hallucination.
- Insight Generation and API Integration
Final outputs can be pushed into enterprise systems (e.g., claims platforms or legal CRMs) via secure APIs.

The doclens.ai platform is not a generic document automation tool—it is a domain-specific AI solution for professionals in insurance, legal, and compliance domains. Through a powerful combination of language models, structured extraction pipelines, and human-aligned evaluation rubrics, the company has positioned itself as a critical enabler of next-generation document intelligence. Its solutions not only save time and reduce errors but also elevate the strategic role of AI in decision-making within highly regulated industries.

CHAPTER 3

SCOPE OF WORK

During my Data Science Spring Internship at doclens.ai, I had the opportunity to work on two high-impact, research-driven projects that lie at the heart of the company's core offerings. These projects involved cutting-edge technologies, real-world legal data, and complex document processing pipelines aimed at enhancing the functionality, accuracy, and reliability of AI-based systems in regulated environments.

The two primary projects assigned to me during this internship were:

1. Rubric-Based Evaluation of Generative AI Responses in Insurance and Legal Domains
2. Automated Legal Case Summarization Using Public PACER Records Versus LexisNexis References

Together, these projects defined the technical and analytical scope of my internship. They exposed me to a wide range of tasks including rubric design, large language model (LLM) evaluation, legal document preprocessing, prompt engineering, structured information extraction, and comparative analysis. The scope also required me to work collaboratively with engineers and legal experts while ensuring that all findings were aligned with the broader goals of responsible AI development and domain-specific utility.

1. Rubric-Based Evaluation and Reasoning Analysis

This project addressed a critical challenge at the intersection of generative AI and insurance claims processing—how to accurately evaluate and trust the responses generated by language models when used in professional workflows involving legal and financial implications. doclens.ai sought to improve upon existing evaluation frameworks, particularly those that were too generic or insufficiently tailored to domain-specific use cases. My work in this project focused on three major areas: rubric design, model response analysis, and evaluation metrics development.

a) Rubric Design and Customization for Insurance Evaluation

The first task involved studying the current limitations of existing frameworks such as RAGAS (Retrieval-Augmented Generation Assessment). These frameworks, while useful for general-purpose evaluation, lacked the depth and specificity needed to assess responses generated for insurance-related prompts involving bodily injury claims, liability questions, and policy interpretations.

To address this, I collaborated with the Head of AI and legal advisors to develop a domain-specific evaluation rubric. This rubric included customized categories such as:

- Factual Accuracy
- Reasoning Completeness
- Legal and Contextual Alignment
- Hallucination Rate
- Answer Structure and Readability

Each category was given a clearly defined scoring system, ranging from binary TP/FP/FN tagging to 5-point scale ratings based on qualitative criteria. This structured rubric allowed for more granular evaluation of AI-generated outputs and facilitated the development of more accurate performance dashboards.

b) Model Output Tagging and Comparative Analysis

Next, I was responsible for evaluating and comparing model outputs from multiple large language models, including Claude 3 Haiku, Claude 3.5 Sonnet, and DeepSeek R1. These models were tasked with responding to prompts related to insurance claims analysis. For each generated response, I conducted a detailed manual tagging exercise, identifying true positives (correctly answered details), false positives (fabricated or unsupported assertions), and false negatives (missed but critical information).

This comparison required not only technical skill but also an understanding of the legal and domain-specific context, as many responses included nuanced phrases about claim causality, injury descriptions, or legal liability, which had to be judged against known ground truths.

c) Preprocessing and Fairness Optimization

One major challenge in model evaluation was the issue of superficial mismatches. Many model responses were penalized due to formatting differences, punctuation errors, or variations in entity representation, even when the actual content was correct. To address this, I designed and implemented a preprocessing routine that standardized outputs by removing extra white space, aligning punctuation, normalizing currency formats, and ensuring consistent case references.

As a result of this preprocessing, the evaluation became significantly more accurate. We recorded a measurable 8.1% increase in precision and reduced false misclassifications, which previously skewed the scoring.

d) Metric Development and Result Synthesis

Using the tagged outputs and rubric scores, I generated evaluation matrices and performance dashboards showing precision, recall, and F1 scores across various models and prompt types. This quantitative analysis helped reveal the following insights:

- Rubric modifications significantly improved the score distribution of Claude 3 Haiku, making it more sensitive to reasoning quality.
- Claude 3.5 Sonnet remained rigid and frequently under-scored valid responses, suggesting its utility was better suited to high-stakes or conservative use cases.
- DeepSeek R1 demonstrated superior contextual understanding and provided more explainable evaluation metrics due to its reasoning capabilities.

2. Legal Case Summarization: PACER versus LexisNexis

The second major project I worked on involved evaluating the capability of modern LLMs to generate legally useful summaries from public court records (PACER) and comparing these outputs to LexisNexis summaries, which are often used by legal professionals for quick case reference. This project aimed to determine whether PACER documents, despite being unstructured and raw, could serve as a cost-effective and open-source alternative for generating accurate legal summaries.

a) Data Collection, Classification, and Preprocessing

My first responsibility was to organize and curate a dataset of legal case files. This included:

- Identifying and grouping all PACER documents relevant to four major cases.
- Retrieving their corresponding LexisNexis summaries.
- Performing OCR conversion where scanned PDFs were present.
- Classifying over 20 types of legal documents (e.g., Complaints, Motions, Judgments, Affidavits).
- Each document was labeled and indexed to facilitate structured information extraction and case-wise comparison.

b) Prompt Development for Information Extraction

I developed structured prompts designed to extract over 30 specific fields from each PACER document, including:

- Plaintiff and defendant names
- Court and judge details
- Injury descriptions and award amounts

- Case summary (3–4 paragraph version)
- Attorney contact information
- Practice area, trial duration, and jury deliberation

These prompts were tested across ChatGPT-4 Turbo and DeepSeek R1, and their performance was recorded.

c) Structured vs Freeform Output Analysis

While both models were able to return responses, the results varied in quality and completeness. Key observations included:

- ChatGPT-4 Turbo tended to simplify complex legal facts and often omitted key details such as trial outcomes or specific injuries.
- DeepSeek R1, though more concise, captured field-specific details with higher fidelity, especially when explicitly prompted.

I documented these differences in a field-wise accuracy matrix, analyzing strengths and gaps in each model's response.

d) Summary Quality Comparison

I conducted a comparative analysis of PACER-derived summaries and LexisNexis summaries on the following dimensions:

- Legal completeness
- Structural coherence
- Fact inclusion and factual hallucination
- Use of legal terminology

The results demonstrated that while PACER-based summaries had potential, they often lacked context or required extensive model intervention. LexisNexis summaries, being professionally curated, remained more complete and reliable. However, with sufficient preprocessing and follow-up prompting, LLMs were able to generate summaries that were reasonably close in fidelity—especially for use cases involving third-party review or non-critical legal research.

3. Collaborative and Iterative Development

Throughout the course of these projects, I had the opportunity to engage in collaborative efforts with doclens.ai's cross-functional teams. This included:

- Participating in weekly meetings.
- Presenting findings to the Head of AI and Chief Product Officer.
- Sharing feedback loops with fellow interns and data scientists.
- Documenting the entire evaluation process for reuse by future evaluators.

- This collaboration helped reinforce the iterative nature of product development and the importance of cross-domain feedback in designing AI systems for specialized industries like law and insurance.

The scope of work undertaken during my internship at doclens.ai was diverse, intellectually challenging, and immensely rewarding. It exposed me to real-world constraints, cross-domain problem-solving, and the intricacies of deploying AI solutions in sensitive, regulated environments. From building custom rubrics and evaluating LLM outputs to structuring unformatted legal records and drawing comparative performance metrics, every aspect of the work contributed to a deeper understanding of AI model evaluation, legal documentation, and responsible automation.

This experience not only broadened my technical skill set but also instilled a deeper appreciation for the importance of trust, accuracy, and transparency in the deployment of generative AI technologies in domains that directly impact human lives and legal outcomes.

CHAPTER 4

LEARNING OBJECTIVES

The internship at doclens.ai provided an intellectually stimulating environment where theoretical knowledge intersected with real-world applications in the fields of data science, artificial intelligence, legal technology, and insurance analytics. The structured mentorship, challenging projects, and exposure to industry-grade tools and workflows significantly contributed to both my technical and professional development. This chapter outlines the key learning objectives that were achieved through hands-on tasks, team collaboration, research-oriented thinking, and applied problem-solving.

1. Understanding Generative AI Evaluation in Regulated Domains

- One of the foremost learning goals during the internship was to understand how generative AI models—particularly large language models (LLMs)—can be assessed for accuracy, reliability, and contextual relevance in sensitive industries like insurance and law. This involved:
- Studying the limitations of off-the-shelf evaluation tools and recognizing the inadequacies of generic scoring systems in domain-specific contexts.
- Learning to design customized evaluation rubrics tailored for insurance claims, legal liability, and injury-related content.
- Understanding how to measure and interpret key performance indicators such as True Positives (TP), False Positives (FP), and False Negatives (FN) within the rubric evaluation process.
- Gaining exposure to comparative analysis using multiple models (Claude 3, ChatGPT-4, DeepSeek R1) and observing their behavior across prompt types.
- This learning enabled me to appreciate the nuance of evaluating AI systems where factual correctness and reasoning integrity are non-negotiable.

2. Gaining Proficiency in Prompt Engineering for LLMs

- Another major learning objective was mastering prompt engineering, the art and science of designing input instructions that elicit accurate, relevant, and structured responses from language models.
- I learned how to:
- Construct effective baseline prompts for structured data extraction.
- Apply few-shot prompting techniques to guide model responses using contextual

examples.

- Design follow-up prompts to improve summarization depth or correct underperformance in initial outputs.
- Analyze how small variations in phrasing or format can significantly alter model behavior, especially in terms of hallucination, verbosity, or detail fidelity.
- By the end of the internship, I was able to strategically craft prompts that enhanced the performance of models for use in document summarization, field-level extraction, and scoring.

3. Exploring Legal and Insurance Document Workflows

- Working with PACER court documents and LexisNexis case summaries allowed me to deeply engage with legal document structures and insurance terminology, which are often complex and semantically dense.
- Key learnings included:
- Understanding how legal case files are organized, from Complaints and Motions to Verdicts and Affidavits.
- Identifying the different types of information embedded in documents (e.g., injuries, court dates, attorney names, trial outcomes).
- Gaining familiarity with terminology related to bodily injury claims, risk assessments, and legal precedents.
- Observing how inconsistencies in document formatting or scan quality affect downstream data extraction and model interpretation.
- This experience contributed to a practical understanding of document classification and content curation in legal and insurance workflows, which is critical when building AI products for these domains.

4. Designing and Implementing Data Preprocessing Pipelines

- A recurring challenge throughout the internship was dealing with inconsistencies in raw data, especially in AI model evaluation and court document analysis. I developed preprocessing methods to:
- Normalize extracted text for whitespace, punctuation, and entity naming conventions.
- Standardize currency formats, legal citations, and document structures to reduce variability in model comparison.
- Improve the robustness of evaluation scripts and scoring workflows by mitigating

superficial mismatches that could skew results.

- These efforts helped develop my skills in data cleaning, text normalization, and pipeline debugging, which are foundational competencies in any data-intensive AI application.

5. Contribute to Real-World AI-Driven Products

- Participate in the ongoing development of ClaimLens, the company's AI-powered Insurance Risk Assistant.
- Generate and benchmark summaries for real PACER case files and evaluate LLM performance in production-like conditions.
- Work alongside mentors and team members to improve response accuracy, reduce hallucination, and build evaluation-ready datasets.
- Understand how AI tools support business workflows in claim triaging, legal risk assessment, and case prioritization.

CHAPTER 5

CHALLENGES AND SOLUTION

During the internship at DocLens.ai, I encountered a variety of technical and domain-specific challenges while working with Large Language Models (LLMs), document evaluation pipelines, and legal data. These challenges provided valuable learning opportunities and shaped the approach toward building more reliable, accurate, and scalable AI systems. The following outlines the key obstacles faced during different phases of the internship and the solutions implemented to address them.

1. Handling Sensitive Legal and Insurance Data

Challenge:

The evaluation of LLM-generated responses required precise alignment between the user input, the model response, and the ground truth summary. Many of the legal and insurance datasets—especially those derived from PACER case files—contained unstructured, lengthy, and inconsistently formatted text. This made it difficult to compare LLM outputs fairly and effectively using automated metrics.

Solution:

A structured preprocessing pipeline was implemented to clean and reformat the datasets while preserving their semantic content. The primary goal was to ensure that inputs passed to the RAG pipeline were consistent and that the outputs could be reliably evaluated using tools like RAGAS.

2. Incompatibility of DeepSeek with LangChain and RAGAS

Challenge:

Among the models tested, DeepSeek showed strong reasoning capabilities, particularly in extracting and synthesizing key facts from complex insurance queries. However, its output structure did not always conform to the formatting expected by automated tools like RAGAS, which are sensitive to structural alignment for scoring.

Solution:

Given the model's qualitative strength and stable performance in reasoning-heavy scenarios, it was excluded from RAGAS-based evaluation. Instead, DeepSeek's outputs were reviewed manually using domain-specific rubrics that prioritized accuracy, legal tone, and factual completeness. This allowed for a fairer and more insightful comparison that respected the model's inherent output style.

3. Conservative Scoring by Claude Sonnet

Challenge:

Claude Sonnet, despite generating contextually correct responses, consistently received lower metric scores in RAGAS evaluations. The model often used verbose or slightly rephrased language that caused mismatches with the expected summary format, leading to unfair penalties in automated scoring.

Solution:

To address this issue, the evaluation was transitioned to Claude Haiku, which produced more concise and structured outputs aligned with the formatting assumptions of RAGAS. Additionally, rubrics and metric thresholds were refined to account for linguistic variation without compromising on factual accuracy. This shift improved both the reliability of scoring and the interpretability of model comparisons.

5. Hallucination and Irrelevant Outputs in LLM Responses

Challenge:

Some LLMs occasionally produced hallucinated content—fabricated facts or overly generic responses that were not supported by the source material.

Solution:

Few-shot prompting and prompt engineering techniques were employed to improve context awareness. Retrieval mechanisms using RAG pipelines were fine-tuned to ensure that the context fed into the LLM was highly relevant and domain-specific.

6. Designing Domain-Specific Evaluation Rubrics

Challenge:

Standard evaluation metrics did not fully capture the nuances of legal and insurance-based responses. Evaluating factual accuracy and legal reasoning required specialized criteria.

Solution:

A custom rubric framework was developed, including dimensions such as liability interpretation, factual consistency, legal tone, and summary coherence. These rubrics were refined through mentor feedback and applied in model comparisons across datasets.

7. Managing PACER File Complexity

Challenge:

PACER documents were long, unstructured, and varied significantly in format. Summarizing them accurately using LLMs required consistent formatting and cleaning

before input.

Solution:

A preprocessing pipeline was created to extract key sections of legal rulings and standardize input formats for summarization. This improved the relevance and coherence of LLM-generated summaries, especially when benchmarking against sources like LexisNexis.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

Operating System: Windows 10/11

Development Environment:

- Visual Studio Code
- Git
- Jupyter Notebook

Programming Languages

- Language: Python 3.10+

Frameworks/Libraries:

- LangChain
- RAGAS
- HuggingFace Transformers
- Pydantic
- Pandas

Cloud Platforms:

- Amazon Web Services – Bedrock

Hardware requirements:

- Processor: Intel i5 or AMD Ryzen 5 and above
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: At least 256 GB SSD
- Internet: Stable high-speed connection

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

During my internship at DocLens.ai, I had the opportunity to contribute to real-world projects that sit at the intersection of artificial intelligence, legal document processing, and insurance risk evaluation. The work was divided across three distinct project phases, each with its own goals and technical challenges. These experiences helped me strengthen my understanding of LLMs, RAG pipelines, and AI evaluation workflows, while allowing me to actively contribute to a production-adjacent AI system.

1. Learning and Research Contributions (Phase 1)

In the initial phase of the internship, I focused on building foundational knowledge required to contribute to AI-driven document evaluation tasks. Key accomplishments in this phase include:

- Completed DeepLearning.AI's short modules on:
 - Transformer architecture
 - Semantic search and retrieval
 - Advanced techniques in Retrieval-Augmented Generation (RAG)
- Reviewed LangChain documentation and tested core modules such as prompt templates, retrievers, and chains.
- Understood the role of grounding, context windows, and token limitations in AI document summarization.

These efforts helped build the necessary technical background to work effectively with the tools and models used in later phases.

2. Model Evaluation and Prompt Design (Phase 2)

This phase was focused on evaluating LLM-generated responses to auto insurance-related user queries, using both manual rubrics and automated metrics. My contributions included:

- Developed domain-specific scoring rubrics to assess factual accuracy, relevance, and tone of responses.
- Evaluated outputs from multiple LLMs (Claude Haiku, Claude Sonnet, DeepSeek) against expert-verified ground truth summaries.
- Preprocessed raw datasets to ensure consistent formatting between user input, model output, and reference summaries.

- Conducted comparative evaluation of model performance:
- DeepSeek was manually evaluated for strong reasoning but excluded from automated scoring due to formatting deviations.
- Claude Sonnet produced valid responses but underperformed on RAGAS metrics, prompting a switch to Claude Haiku, which produced more structured and concise outputs.
- Refined prompt formats and experimented with few-shot prompting to reduce hallucinations and improve model coherence.

This work played a direct role in testing the quality and reliability of LLMs within an insurance-specific domain.

3. Legal Document Summarization (Phase 3)

As part of the final and ongoing phase of the internship, I was involved in summarizing legal documents obtained from PACER (Public Access to Court Electronic Records) to support insurance litigation analysis. Key contributions:

- Parsed and cleaned legal documents for use in summarization pipelines.
- Designed prompts to generate concise case summaries from lengthy rulings.
- Benchmarked generated outputs against legal references from trusted sources like LexisNexis.
- Identified performance issues in long-context summarization and proposed improvements using chunking and context prioritization.
- Collaborated with the AI team to document best practices for legal summarization using LLMs.

This phase allowed me to apply my prior learning in a real-world setting and contribute directly to DocLens.ai's ClaimLens product workflows.

CHAPTER 7

REFLECTION AND ANALYSIS

The internship at DocLens.ai was a valuable learning experience that allowed me to apply AI concepts in a practical, domain-specific setting. Being part of projects that involved legal document summarization and the evaluation of large language models (LLMs) exposed me to real-world challenges in integrating generative AI into high-stakes workflows. Working with tools such as LangChain, RAGAS, and Amazon Bedrock, I developed a deeper understanding of retrieval-augmented generation, prompt structuring, and model orchestration within insurance and legal use cases.

One of the most rewarding aspects of the internship was the opportunity to evaluate model outputs in a domain where accuracy and legal interpretation are critical. While DeepSeek was excluded from automated scoring due to formatting differences, it performed well in reasoning, and I learned how to adapt manual rubrics to measure its effectiveness. In contrast, Claude Sonnet, though accurate, often received lower scores from RAGAS due to output phrasing. This led to a shift to Claude Haiku and subsequent refinements in evaluation criteria. These challenges taught me the importance of aligning both tools and expectations when assessing LLM outputs in sensitive domains.

The internship also strengthened my analytical and problem-solving skills. I was able to identify gaps in existing evaluation workflows, iterate on prompt formats, and participate in data preprocessing to improve model input consistency. Working through ambiguity and debugging edge cases gave me a better appreciation for how small adjustments can significantly improve output quality. Regular mentor feedback sessions helped guide this process and encouraged a structured, detail-oriented approach to experimentation.

Beyond technical learning, I gained exposure to collaborative development in a hybrid work environment. Interactions with mentors and peer interns enhanced my communication and project planning skills. I also learned the value of documenting workflows, managing version control, and aligning with agile feedback cycles. Overall, this internship has not only improved my technical confidence but also reinforced my interest in building responsible and effective AI systems for complex domains like insurance and legal tech.

CHAPTER 8

RECOMMENDATION

1. Introduce Well-Defined Internship Structures

DocLens.ai can enhance future internship experiences by adopting a structured framework that sets clear expectations from the beginning. Clearly defining roles, learning goals, timelines, and deliverables will help interns stay focused and aligned with team objectives. A balanced schedule with guided sessions and task-based learning can accelerate skill development while enabling interns to track their growth more effectively.

2. Foster Academic Collaborations

Building partnerships with academic institutions can help bridge the gap between theoretical learning and industry application. Collaborations through guest lectures, workshops can expose interns to real-world challenges in areas such as insurance automation, legal-tech, and generative AI evaluation.

3. Establish Strong Mentorship Channels

Assigning each intern a dedicated mentor who provides technical guidance and regular feedback can significantly improve the learning experience. Scheduled mentor–intern check-ins, project reviews, and open feedback loops can help interns better navigate tools, workflows, and evaluation methodologies with greater confidence.

4. Encourage Active Participation and Creative Problem Solving

Providing opportunities for interns to share ideas, attend team meetings, and contribute to mini-projects promotes a sense of ownership and innovation. This not only boosts intern engagement but also introduces new perspectives that can enhance ongoing product development and experimentation.

5. Create Networking and Exposure Opportunities

Facilitating interaction with cross-functional teams—including AI researchers, product managers, and legal consultants—can help interns gain a broader understanding of how AI solutions are developed and applied across domains. Internal demo days, AMA sessions, or tech talks would provide valuable industry exposure beyond the core internship scope.

CHAPTER 9

CONCLUSION

The ongoing internship at DocLens.ai has provided a deeply engaging and intellectually rewarding experience, offering both technical learning and real-world exposure to AI applications in the InsurTech domain. Working at the intersection of artificial intelligence, legal documentation, and insurance risk evaluation has significantly broadened my understanding of how advanced technologies can be integrated into sensitive, high-stakes environments.

Over the course of this internship, I have gained hands-on experience with industry-standard tools such as LangChain, Amazon Bedrock, and Python-based evaluation frameworks. I have contributed to multiple project phases, from developing rubric-based evaluation strategies for LLM outputs to summarizing real-world legal documents like PACER court files. Each task has helped reinforce key AI concepts such as Retrieval-Augmented Generation (RAG), prompt engineering, and domain-specific response validation.

This internship has not only helped me strengthen my technical skill set but also nurtured soft skills like communication, collaboration, and agile problem-solving. Regular mentorship and feedback sessions enabled me to align my work with industry standards while also encouraging experimentation and critical thinking.

Though the internship is still ongoing, the learning so far has been both impactful and motivating. The experience at DocLens.ai has reaffirmed my interest in working with applied AI systems and has provided a solid foundation to pursue future opportunities in the fields of legal-tech, InsurTech, and AI model evaluation.

Moreover, I got to see how mentorship and guidance from experienced team members can make a huge difference. Regular feedback and review sessions pushed me to think critically and improve with every task. The internship not only gave me technical confidence but also helped me understand the importance of communication, problem-solving, and team spirit in a work setting.

In conclusion, my time at DocLens.ai has been a valuable stepping stone in my career journey. It has boosted my confidence as a developer, given me practical skills, and taught me how to adapt, learn, and contribute in a fast-paced tech environment. I am truly grateful for this experience and excited to apply what I've learned to future opportunities.

CHAPTER 10

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